

Question number	Answer	Additional guidance	Mark
8(a)(i)	anthropogenic (climate change / activity / events)		(1)

Question number	Answer	Additional guidance	Mark
8(a)(ii)	An explanation that includes the following points: • because {there is no evidence that climate change has actually caused the forest fires / it is only a correlation} (1) • because certain {parties / politicians / business people} {have different opinions / want to continue their activities} (1) • forest fires are caused for other reasons / other (non-human) factors cause climate change (1) • lack of education (1)	ACCEPT lack of trust in scientists ACCEPT named factor e.g. volcanic eruptions, solar flares	(2)

Question number	Answer	Mark
8(b)	Local atmosphere: • increase in CO₂ levels in the atmosphere • increase global warming • increase in soot / smoke • when the organic matter is burnt • soot prevents so much light penetrating • details of global warming • extent depends on {size / frequency} of fires • less CO₂ being removed from the atmosphere • as there are fewer trees to photosynthesise • CO₂ levels in atmosphere will begin to drop • as CO₂ diffuses away from the area • new plants will begin to grow • growing plants will be photosynthesising faster (than established plants) • therefore will be absorbing CO₂ from the atmosphere Local biodiversity: • biodiversity will decrease • because plants and animals are killed • because plant and animal homes will be destroyed • animals depend on plants • effect on food chains • fragmentation of habitat • fewer organisms of a species could affect gene pool • increase in inbreeding further decreasing biodiversity • plants will begin to sprout • seeds will germinate • (secondary) succession begins • increasing biodiversity • as biodiversity of plants increases so does the biodiversity of animals • eventually climax community re-established	(6)

			Additional guidance
vel 0	0	No awardable content	
vel 1	1-2	Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.	Simple links made between the forest fires and the effects on biodiversity and atmosphere 1 mark = effect of forest fires on either biodiversity or atmosphere 2 marks = effect of forest fires on both biodiversity and atmosphere
vel 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.	Extended links made the forest fires and the effects on biodiversity and atmosphere 3 marks = comments on both biodiversity and atmosphere with extended comment on one 4 marks = extended comments made on both biodiversity and atmosphere
vel 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant biological facts / concepts. Consequences are discussed which supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.	Clear discussions on the short-term and long-term effects of the forest fires on biodiversity and atmosphere 5 marks = plus clear discussion on the short-term and long-term effects of the forest fires on either biodiversity or atmosphere 6 marks = plus clear discussions on the short-term and long-term effects of the forest fires on both biodiversity and atmosphere

Question number	Answer	Additional guidance	Mark
(c)	An explanation that includes four of the following points: - choose different trees if original trees not suited to the new conditions (1) - credit one change to consider (1) - credit second change to consider (1) - credit an explanation for one of the choices made (1) - consideration of {needs of / needs for} animals (1)	ACCEPT keep same trees if they are suited to new conditions Choose trees that are adapted to the new conditions ACCEPT increased light intensity if linked to lack of tree cover decreases mineral ions if linked to leaching e.g. fire resistant so less likely to be destroyed again higher productivity {so habitat replaced faster / <u>more</u> carbon dioxide removed from the atmosphere} (drought) so that they will not dehydrate (flooding) so that {ground will not be water-logged / trees do not get washed away (temperature) so that their enzymes will function	(4)